

Test conditions for the measurement or calculation of harmonic currents are given in Clause 7.

4.2.3 Repeatability

The repeatability of the average value for the individual harmonic currents over the entire test observation period shall be better than $\pm 5\%$ of the applicable limit, when the following conditions are met:

- the same equipment under test (EUT) (not another of the same type, however similar);
- identical test conditions;
- the same test system;
- identical climatic conditions, if relevant.

NOTE This repeatability requirement serves the purpose of defining the necessary test observation period, see 4.2.7. It is not intended to serve as a pass/fail criterion for the assessment of compliance with the requirements of this standard.

4.2.4 Starting and stopping

When a piece of equipment is brought into operation or is taken out of operation, manually or automatically, harmonic currents are not taken into account for the first 10 s, or until the equipment is fully in or out of operation, whichever is longer, following the switching event.

The equipment under test shall not be in stand-by mode (see 3.15) for more than 10 % of any observation period.

4.2.5 Application of limits

The average value for the individual harmonic currents, taken over the entire test observation period shall be less than or equal to the applicable limits in Tables 2 to 5.

For each harmonic order, all 1,5 s smoothed r.m.s. harmonic current values, as defined in 4.2.2, shall be less than or equal to 150 % of the applicable limits.

For the calculation of *THC* and *PWHC*, individual harmonic currents below 1 % of the reference current are disregarded.

4.2.6 Test report

The test report may be based on information supplied by the manufacturer to a testing facility, or be a document recording details of the manufacturer's own tests. It shall include all relevant information for the test conditions, the test observation period and the determination of the reference current showing compliance with the present standard.

The test report shall include:

- the values of the input current measured and specified by the manufacturer for the determination of the reference current I_{ref} , according to 4.1;
- the short circuit ratio used for calculation or test;
- the required minimum short circuit ratio;
- and a statement about the table applied (i.e. about the type of equipment).

4.2.7 Test observation period

Observation periods (T_{obs}) for four different types of equipment behavior are considered and described in Table 1.

Table 1 – Values of the observation period

Type of equipment behavior	Observation period
Quasi-stationary	T_{obs} of sufficient duration to meet the requirements for repeatability in 4.2.3.
Short cyclic ($T_{\text{cycle}} \leq 2,5 \text{ min}$)	$T_{\text{obs}} \geq 10$ cycles (reference method) or T_{obs} of sufficient duration or synchronization to meet the requirements for repeatability in 4.2.3. ^a
Random	T_{obs} of sufficient duration to meet the requirements for repeatability in 4.2.3.
Long cyclic ($T_{\text{cycle}} > 2,5 \text{ min}$)	Full equipment program cycle (reference method) or a representative 2,5 min period considered by the manufacturer as the operating period with the highest <i>THC</i> .
^a By synchronization is meant that the total observation period is sufficiently close to including an exact integral number of equipment cycles in such a way that the requirements for repeatability in 4.2.3 are met.	

4.3 Equipment consisting of several self-contained items

Where individual self-contained items of equipment (possibly, but not necessarily, of different manufacture) are assembled in a rack or case, compliance with the present standard shall be achieved either for the system as a whole or for each individual self-contained item at the manufacturer's discretion.

5 Requirements and limits for equipment

5.1 Control methods

Only symmetrical control methods (see IEC 60050:1990, 161-07-11) are allowed under normal operating conditions.

Symmetrical control methods which are not multicycle control (see IEC 60050:1990, 161-07-05) and which are used for the control of the power supplied to heating elements are only allowed for professional equipment whose primary purpose considered as a whole is not for heating. In addition, all the three following conditions apply:

- a) the relevant limits are not exceeded when tested at the supply input terminals;
- b) it is necessary to control precisely the temperature of a heater whose thermal time constant is less than 2 s;
- c) there is no other technique economically available.

NOTE For the purposes of this standard, burst firing is deemed to be symmetrical multicycle control.

5.2 Limits for emission

The limits given apply to 230/400 V, 50 Hz systems. The limits for the other systems will be added in a future edition of this standard.

NOTE 1 In some non-European countries, the proposed methodology cannot be applied because the short-circuit power data is not always available.

The harmonic current limits specified in the tables apply to each of the line currents and not to current in the neutral conductor.

For equipment with multiple rated currents, an assessment is made for each current.

As an example (for the same equipment):

Rated voltage: 230 V single phase, rated current: x A per phase, assessment and test at 230 V.

Rated voltage: 400 V three phase, rated current: y A per phase, assessment and test at 400 V.

The harmonic current limits are specified in Tables 2 to 5.

Equipment complying with the harmonic current emission limits corresponding to $R_{sce} = 33$ is suitable for connection at any point of the supply system.

NOTE 2 Values are based on a minimum value of $R_{sce} = 33$. Short-circuit ratios less than 33 are not considered.

NOTE 3 In order to reduce the depth of commutation notches of converters, a short-circuit ratio higher than 33 may be necessary.

For equipment not complying with the harmonic current emission limits corresponding to $R_{sce} = 33$, higher emission values are allowed, under the assumption that the short-circuit ratio R_{sce} is greater than 33. It is expected that this will apply to the majority of equipment with input current above 16 A per phase. See requirement for product documentation in Clause 6.

Table 2 is applied to equipment other than balanced three-phase equipment and Tables 3, 4 and 5 are applied to balanced three-phase equipment.

Table 3 may be used for any balanced three-phase piece of equipment.

Table 4 may be used with balanced three-phase equipment if any one of ~~these~~ the following conditions is met:

- a) The 5th and 7th harmonic currents are each less than 5 % of the reference current during the whole test observation period.

NOTE 4 This condition is normally fulfilled by 12-pulse pieces of equipment.

- b) The design of the piece of equipment is such that the phase angle of the 5th harmonic current has no preferential value over time during normal operation and can take any value ~~in the whole interval $[0^\circ, 360^\circ]$~~ from 0° to 360° .

NOTE 5 This condition is normally fulfilled by converters with fully controlled thyristor bridges.

- c) Among all the 5th harmonic current values measured in each DFT time window as defined in IEC 61000-4-7 over the entire test observation period, the phase angle of the 5th harmonic current related to the fundamental phase-to-neutral voltage (see 3.16) is in the range of 90° to 150° ~~during the whole test observation period~~ for at least 99 % of the harmonic current values that have an RMS value higher than 10,7 % of the reference current.

NOTE 6 This condition is normally fulfilled by equipment with an uncontrolled rectifier bridge and capacitive filter, including a 3 % ~~a.c.~~ AC or 4 % ~~d.c.~~ DC reactor.

Table 5 may be used with balanced three-phase equipment if any one of ~~these~~ the following conditions is met:

- d) The 5th and 7th harmonic currents are each less than 3 % of the reference current during the whole test observation period.

NOTE 7 This condition is normally fulfilled by 12-pulse pieces of equipment.

- e) The design of the piece of equipment is such that the phase angle of the 5th harmonic current has no preferential value over time during normal operation and can take any value ~~in the whole interval $[0^\circ, 360^\circ]$~~ from 0° to 360° .

NOTE 8 This condition is normally fulfilled by converters with fully controlled thyristor bridges.

- f) Among all the 5th harmonic current values measured in each DFT time window as defined in IEC 61000-4-7 over the entire test observation period, the phase angle of the 5th harmonic current related to the fundamental phase-to-neutral voltage (see 3.16) is in the range of 150° to 210° ~~during the whole test observation period~~ for at least 99 % of the harmonic current values that have an RMS value higher than 10,7 % of the reference current.

NOTE 9 This condition is normally fulfilled by a 6-pulse converter with a small ~~d.c.~~ DC link capacitance, operating as a load.

Table 3, Table 4 or Table 5 can be applied to hybrid equipment in one of the following circumstances:

- a) hybrid equipment having a maximum 3rd harmonic current of less than 5 % of the reference current, or
- b) there is provision in the construction of hybrid equipment to separate the balanced three-phase and the single-phase or interphase loads for the measurement of supply currents, and when the current is being measured, the part of the equipment being measured draws the same current as under normal operating conditions. In that case, the relevant limits shall be applied separately to the single-phase or interphase part and to the balanced three-phase part. Table 3, Table 4 or Table 5 applies to the current of the balanced three-phase part, even if the rated current of the balanced three-phase part is less than or equal to 16 A per phase. Table 2 applies to the current of the single-phase or interphase part, but if the rated current of the single-phase or interphase part is less than or equal to 16 A, the manufacturer may apply the relevant limits of IEC 61000-3-2 to the single-phase or interphase part instead of the limits stated in Table 2.

For verification purposes, when circumstance b) above applies, the manufacturer shall state in the product documentation the rated current and give in the test report the measured and specified values of the input current as defined in 4.1, for each separate load. The value of R_{SCe} for this type of hybrid equipment is determined as follows:

- the minimum R_{Sce} value is first determined for each of the two loads, using the reference current of the considered part for the calculation of the harmonic current emissions to be compared to the limit values given in Tables 2 to 5; in case IEC 61000-3-2 is applied to the single-phase or interphase part instead of Table 2 limits, the minimum R_{Sce} value for this part is deemed to be equal to 33;
- then, for each of the two parts, the minimum value of S_{Sc} is calculated from its minimum R_{Sce} value and its rated current (see 3.11 and 3.14);
- finally, the value of R_{Sce} for the hybrid equipment is determined from the highest of both minimum values of S_{Sc} and the rated apparent power of the whole hybrid equipment.

Table 2 – Current emission limits for equipment other than balanced three-phase equipment

[illegible]

Table 3 – Current emission limits for balanced three-phase equipment

Minimum R_{sce}	Admissible individual harmonic current I_h/I_{ref} ^a %				Admissible harmonic parameters %	
	I_5	I_7	I_{11}	I_{13}	THC/I_{ref}	$PWHC/I_{ref}$
33	10,7	7,2	3,1	2	13	22
66	14	9	5	3	16	25
120	19	12	7	4	22	28
250	31	20	12	7	37	38
≥350	40	25	15	10	48	46
The relative values of even harmonics up to order 12 shall not exceed 16/h %. Even harmonics above order 12 are taken into account in <i>THC</i> and <i>PWHC</i> in the same way as odd order harmonics.						
Linear interpolation between successive R_{sce} values is permitted.						
^a I_{ref} = reference current; I_h = harmonic current component.						

Table 4 – Current emission limits for balanced three-phase equipment under specified conditions (a, b, c)

Minimum R_{sce}	Admissible individual harmonic current I_h/I_{ref} ^a %				Admissible harmonic parameters %	
	I_5	I_7	I_{11}	I_{13}	THC / I_{ref}	$PWHC / I_{ref}$
33	10,7	7,2	3,1	2	13	22
≥ 120	40	25	15	10	48	46
The relative values of even harmonics up to order 12 shall not exceed 16/h %. Even harmonics above order 12 are taken into account in <i>THC</i> and <i>PWHC</i> in the same way as odd order harmonics.						
Linear interpolation between both R_{sce} values is permitted.						
^a I_{ref} = reference current; I_h = harmonic current component.						

Table 5 – Current emission limits for balanced three-phase equipment under specified conditions (d, e, f)

Minimum R_{Sce}	Admissible individual harmonic current I_h/I_{ref} ^a %												Admissible harmonic parameters %	
	I_5	I_7	I_{11}	I_{13}	I_{17}	I_{19}	I_{23}	I_{25}	I_{29}	I_{31}	I_{35}	I_{37}	THC / I_{ref}	$PWHC / I_{ref}$
33	10,7	7,2	3,1	2	2	1,5	1,5	1,5	1	1	1	1	13	22
≥250	25	17,3	12,1	10,7	8,4	7,8	6,8	6,5	5,4	5,2	4,9	4,7	35	70

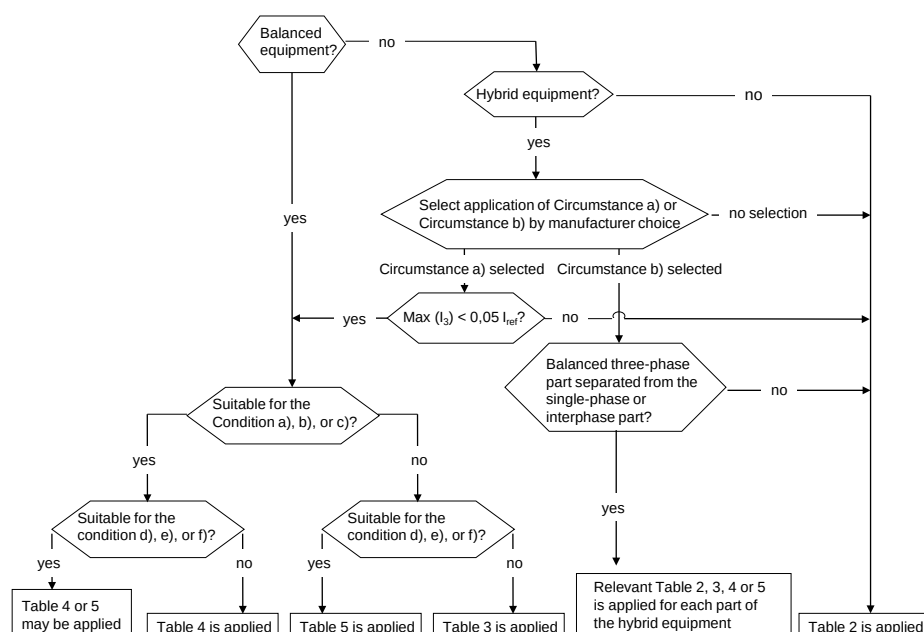
For R_{Sce} equal to 33, the relative values of even harmonics up to order 12 shall not exceed 16/h %. The relative values of all harmonics from I_{14} to I_{40} not listed above shall not exceed 1 % of I_{ref}

For $R_{Sce} \geq 250$, the relative values of even harmonics up to order 12 shall not exceed 16/h %. The relative values of all harmonics from I_{14} to I_{40} not listed above shall not exceed 3 % of I_{ref} .

Linear interpolation between both R_{Sce} values is permitted.

^a I_{ref} = reference current; I_h = harmonic current component.

Figure 3 shows the flowchart of the application procedure of Tables 2 to 5.



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Figure 3 – Flowchart of the application procedure

5.3 Procedure if the reference current is less than 16 A

If equipment that has a rated current above 16 A draws a reference current that is less than or equal to 16 A under the specified test conditions, the manufacturer shall use one of the following procedures:

- Comply with the relevant emission limits given in IEC 61000-3-12, choosing the required R_{Sce} .
- Comply with the absolute limits given in IEC 61000-3-2:2018, Table 1, instead of the proportional limits given in IEC 61000-3-12, using and complying with all other requirements given in IEC 61000-3-12, and the manufacturer shall state in the instruction manual "Equipment complying with IEC 61000-3-12", without having to declare a minimum short circuit power S_{Sc} .
- Change the test observation period to a representative 2,5 min period expected as the operating period with the highest THC , and proceed as in a) above, choosing the required R_{Sce} .

The manufacturer shall state in the test report which of these procedures was used, so that subsequent tests are carried out with the same procedure.

6 Product documentation

For equipment complying with the harmonic current emission limits corresponding to $R_{Sce} = 33$, the manufacturer shall state in the instruction manual supplied with the equipment:

"Equipment complying with IEC 61000-3-12"

~~For equipment not complying with the harmonic current emission limits corresponding to $R_{Sce} = 33$, the manufacturer shall~~

- ~~• determine the minimum value of R_{Sce} for which the limits given in relevant Table 2, Table 3, Table 4 or Table 5 are not exceeded,~~
- ~~• declare the value of the short-circuit power S_{Sc} corresponding to this minimum value of R_{Sce} (see 3.14) in the instruction manual,~~
- ~~• and instruct the user to determine, in consultation with the distribution network operator if necessary, that the equipment is connected only to a supply of that S_{Sc} value or more. For that purpose, the statement in the instruction manual shall be:~~

~~"This equipment complies with IEC 61000-3-12 provided that the short-circuit power S_{Sc} is greater than or equal to xx at the interface point between the user's supply and the public system. It is the responsibility of the installer or user of the equipment to ensure, by consultation with the distribution network operator if necessary, that the equipment is connected only to a supply with a short-circuit power S_{Sc} greater than or equal to xx."~~

~~where xx is the value of S_{Sc} corresponding to the minimum value of R_{Sce} for which the limits given in the relevant Table 2, 3, 4 or 5 are not exceeded.~~

For equipment not complying with the harmonic current emission limits corresponding to $R_{Sce} = 33$, the manufacturer shall determine the minimum value of R_{Sce} for which the limits given in the relevant Table 2, Table 3, Table 4 or Table 5 are not exceeded and either:

- a) declare the value of the short-circuit power S_{Sc} corresponding to this minimum value of R_{Sce} (see 3.14) in the instruction manual, and instruct the user to determine, in consultation with the distribution network operator if necessary, that the equipment is connected only to a supply of that S_{Sc} value or more, or
- b) declare the value of the short-circuit ratio R_{Sce} corresponding to this minimum value of R_{Sce} (see 3.14) in the instruction manual, and instruct the user to determine, in consultation with the distribution network operator if necessary, that the characteristics of the equipment (S_{equ} or I_{equ}) and the supply (S_{Sc}) are such that a short-circuit ratio of that R_{Sce} value or more is achieved.

For item a), the statement in the instruction manual shall be:

"This equipment complies with IEC 61000-3-12 provided that the short-circuit power S_{Sc} is greater than or equal to xx at the interface point between the user's supply and the public system. It is the responsibility of the installer or user of the equipment to ensure, by consultation with the distribution network operator if necessary, that the equipment is connected only to a supply with a short-circuit power S_{Sc} greater than or equal to xx."

where xx is the value of S_{Sc} corresponding to the minimum value of R_{Sce} for which the limits given in the relevant Table 2, Table 3, Table 4 or Table 5 are not exceeded.

For item b), the statement in the instruction manual shall be:

"This equipment complies with IEC 61000-3-12 provided that the short-circuit ratio R_{Sce} is greater than or equal to xx at the interface point between the user's supply and the public system. It is the responsibility of the installer or user of the equipment to ensure, by

consultation with the distribution network operator if necessary, that the short-circuit ratio R_{Sce} is greater than or equal to xx."

where xx is the minimum value of R_{Sce} for which the limits given in the relevant Table 2, Table 3, Table 4 or Table 5 are not exceeded. The manufacturer shall explain how to calculate S_{sc} from R_{Sce} using the relevant formula in 3.14.

7 Test and simulation conditions

7.1 General

Conformity with this standard may be determined by either of the two methods as follows:

- a) direct measurement (see 7.2);
- b) calculation by validated simulation (see 7.3).

When direct testing or simulation is performed by a party other than the equipment manufacturer for the purpose of verifying harmonic emissions of equipment, direct testing or simulation shall be performed using conditions documented in the test report of the manufacturer. Direct tests shall be verified by direct tests and simulations either by review or by conducting new simulations which duplicate the conditions of the simulation performed by the manufacturer.

7.2 Requirements for direct measurement

As a first step in the measurement process, the manufacturer shall choose a trial value of R_{Sce} , (symbol $R_{Sce\ min}$), based on knowledge of the product design, expected to allow the equipment to comply with the requirements in the relevant table.

The supply source shall then meet the following requirements:

- c) the output voltage U shall be the rated voltage of the equipment. In the case of a voltage range, the output voltage shall be a nominal system voltage according to IEC 60038 (for example: 120 V or 230 V for single-phase or 400 V line-line for three-phase);
- d) while the measurements are being made, the output voltage shall be maintained within $\pm 2,0\%$ and the frequency within $\pm 0,5\%$ of the nominal value;
- e) in the case of a three-phase supply, the voltage unbalance shall be less than 50 % of the voltage unbalance compatibility level given in IEC 61000-2-2;
- f) the harmonic ratios of the output voltage U in no-load condition shall not exceed:
 - 1,5 % for harmonic of order 5;
 - 1,25 % for harmonics of order 3 and 7;
 - 0,7 % for harmonic of order 11;
 - 0,6 % for harmonics of order 9 and 13;
 - 0,4 % for even harmonics of order 2 to 10;
 - 0,3 % for harmonics of order 12 and 14 to 40;
- g) for the application of Tables 2 and 3, the impedance of the supply source is such that the R_{Sce} is equal to or higher than $R_{Sce\ min}$, value expected to allow the compliance of the equipment, with possible insertion of reactors.

For the application of Table 4 or 5, the impedance of the supply source is such that the R_{Sce} is equal to or higher than 1,6 times $R_{Sce\ min}$, value expected to allow the compliance of the equipment, with possible insertion of reactors;

NOTE The factor 1,6 is intended to take into account the fact that if an equipment is connected to a supply that gives a higher R_{Sce} value than $R_{Sce\ min}$, the harmonic emission currents increase. An allowance for this is

already included in Tables 2 and 3, so that no further allowance in terms of the value of R_{sce} to be used for testing is considered necessary.

- h) the impedance of the current-sensing part and the wiring is included in the impedance of the supply source.

NOTE The values of impedance and distortion given above have been chosen as a compromise, considering that high quality supplies of very high current capacity are extremely rare.

The repeatability of results, using different supplies, can be very poor with the above-mentioned values of distortion and impedance. The repeatability using the same supply is not so poor. If at all possible, a supply with lower distortion and impedance should be used.

If compliance is not achieved with the trial value R_{scemin} , a higher value of R_{scemin} shall be chosen and the test repeated, until a value of R_{scemin} is found that achieves compliance. This final value shall be used as the 'minimum value of R_{sce} ' in Clause 6.

The requirements for the measurement instrumentation are given in IEC 61000-4-7.

The currents of balanced three-phase equipment may be measured in one of the phases only, but in case of doubt for hybrid equipment, and in any case for unbalanced three-phase equipment, all three phases shall be tested.

For equipment connected to a single phase supply, it is permissible to measure the current in the neutral conductor instead of the current in the line.

Measurements shall be made at the point of connection between the source and the EUT.

NOTE For the assessment of the emissions, see 4.2 and IEC 61000-4-7.

7.3 Requirements for simulation

Assessment of harmonic current emissions and the corresponding minimum value of R_{sce} can be made by computer simulation of the equipment considered. This procedure may be used when the requirements given in 7.2 concerning the supply source cannot be met. In order to validate the results, the following steps shall be performed.

- a) Measurement of the type of equipment under normal laboratory conditions as described in 7.2, with possible higher voltage distortion, ~~provided that~~ is allowed, however the harmonic levels shall not exceed the compatibility levels given in IEC 61000-2-4, class 3. These measurements shall show that the equipment complies with the relevant limits.

The voltage spectrum during the test as well as the supply impedance (value at fundamental frequency, either directly as such or indirectly in terms of short-circuit power, including the impedance of the current-sensing part and the wiring) shall be recorded.

- b) Simulation of the equipment with the manufacturer's software and procedures:

The measured values of the voltage spectrum and supply impedance are taken as input parameters into the simulation. The harmonic currents calculated by this simulation are compared to the results of the measurement under item a). The simulation is considered validated if the results from simulation do not differ from the results of measurement by more than the following values:

- either $\pm 2\%$ of the reference current;
- or $\pm 10\%$;

for each harmonic current ($h \leq 13$), whichever is larger.

NOTE Current technology does not allow simulations to achieve a high degree of accuracy for high order harmonics, so it is impracticable to set tolerance limits for the comparison of simulation and measurement in this case. In developing a comparison between measurement results and simulation results, the manufacturer is encouraged to measure harmonic values up to order 40 and to consider any deviations between the measurement and the simulation results. However, there is no requirement to validate the simulation for harmonic order greater than 13. For the manufacturer to disregard significant deviations above 13 implies a risk that the product in fact does not comply with the limits.

Measured harmonics less than 1 % of the reference current are not compared as part of the validation.

The validation of the simulation need not be repeated for each product in a range of products with rated current within the range 16 A to 75 A based on the same technology. The simulation is considered to be valid if it is validated for one product at or near each end (within the range 16 A to 75 A) of the product range.

- c) The simulation is repeated with a pure sinusoidal, balanced supply voltage and purely inductive impedance.

For the application of Tables 2 and 3, the impedance shall correspond to a R_{sce} higher than or equal to the trial value $R_{sce\ min}$ (see 7.2) expected to allow the compliance of the equipment.

For the application of Table 4 or Table 5, the impedance shall correspond to a R_{sce} higher than or equal to 1,6 times the trial value $R_{sce\ min}$ expected to allow the compliance of the equipment.

The results of this second simulation are considered to be the relevant harmonic currents to obtain the minimum R_{sce} value from Tables 2 to 5. However, if compliance is not achieved with the trial value $R_{sce\ min}$, a higher value of $R_{sce\ min}$ shall be chosen and the simulation repeated, until a value of $R_{sce\ min}$ is found that achieves compliance. This final value shall be used as the 'minimum value of R_{sce} ' in Clause 6.

7.4 General conditions for test and simulation

Emission tests shall be conducted with the user's operation controls or automatic programs set to the mode expected to produce the maximum total harmonic current (*THC*) under normal operating conditions. This defines the equipment set-up during emission tests and not a requirement to conduct searches for worst-case emissions.

The equipment is tested as presented by the manufacturer. Preliminary operation of motor drives by the manufacturer may be needed before the tests are undertaken to ensure that results correspond to normal use.

NOTE Specific test conditions for the measurement or the simulation assessment of harmonic currents associated with some types of equipment, written in accordance with IEC Guide 107, may be given in the relevant product standards.

Test conditions for some types of equipment are given in Annex A.

The test conditions in Annex C of IEC 61000-3-2 may be applied to other equipment of the relevant types which fall within the scope of this part of IEC 61000.

Annex A (normative)

Type test conditions

A.1 General

The test conditions for the evaluation of harmonic currents associated with some types of equipment are given in the following clauses. For equipment not specifically defined, the general procedure given in 7.4 shall be applied.

Equipment with regenerative capabilities shall be tested in operating conditions where the equipment is a power consumer, i.e. while there is positive power flow, unless the equipment is intended to be used to feed power back into the public supply for more than 20 % of the time in normal operating conditions.

A.2 Test conditions for air conditioners and heat pumps

If the input power of the ~~air conditioner~~ compressor motor or fan of the equipment is controlled by an electronic device so that the revolution speed of the compressor motor or fan is changed in order to get the suitable air temperature, the harmonic currents are evaluated after the operation becomes steady-state in accordance with one of the two methods given below.

For ~~air conditioners~~ equipment having electronically supplied compressors ~~(VSD)~~ and also compressors directly connected to the LV network with a predominantly linear current pattern, the manufacturer may choose to conduct measurements without having the directly connected compressor active. The harmonic current component I_h of the electronically supplied part shall be used for the limit comparison calculations.

The temperature control shall be set to the lowest temperature in the cooling mode and to the highest temperature in the heating mode.

One of the two following methods shall be used.

- a) The ambient temperature shall be at a level resulting in greater than 90 % of the maximum *THC* condition for the ~~air conditioner~~ equipment under normal operating conditions, or a special test mode shall be used which results in a load condition exceeding 90 % of the maximum *THC*. This special test mode may bypass the normal control settings as used by the general public, and allows the testing authority to operate the equipment under test at or near maximum rated current, thereby ignoring the intended function, such as temperature control within a predefined range for air conditioner/heater units. If this special test mode is used, it shall be so documented in the test report, along with the applicable environmental conditions.
- b) If the achievable ambient temperature is such that the maximum *THC* condition cannot be reached, and the special test mode mentioned above is not available or practical, the manufacturer shall specify a test temperature that results in a *THC* of greater than 70 % of the maximum *THC*. If this method is used, the manufacturer shall provide harmonic and reference current data evaluated under the same conditions and at the same test temperature ± 2 °C. The harmonic current evaluation made at the specified test temperature for compliance purposes and the manufacturer's evaluation data obtained at the same test temperature shall meet the requirements of 4.2.5. The requirements of 4.2.5 shall be met and the harmonic limits shall not be exceeded during the manufacturer's in-house test at both the maximum *THC* condition and the specified test temperature resulting in emissions greater than 70 % of the maximum *THC*. The manufacturer's

evaluation data for both the tests at greater than 70 % of the maximum *THC* and at maximum *THC* shall be included in the test report.

~~Air-conditioning systems~~ Equipment having only a compressor motor of the directly connected type have a predominantly linear current pattern. Therefore, they are deemed to comply without testing provided that both of the conditions below are met.

- The compressor motor is an induction motor and its power is supplied directly via a relay, solid-state relay, or mechanical switch with conventional on and off function, with a cycle time of at least 1 min.
- The input current of the directly connected compressor motor and directly connected induction fan motor, if any, exceeds 90 % of the rated current.

However, in case of doubt, the result of a full compliance test according to Clauses 4, 5 and 7 takes precedence over this simplified condition.

A.3 Test conditions for instantaneous water heaters

Measurements of the harmonic currents need to be made from lowest possible controllable input power up to the maximum controllable input power, without switching on an uncontrolled resistive load directly connected to the mains.

If all heating elements can be controlled by power electronics or the value of the maximum controllable input power is not known, the full power range is investigated in 20 approximately equally distributed steps (step size is 5 % of total power).

If the full controllable power range is known and only controllable input power is investigated, this range shall be stated in the test report. This controllable power range is investigated with the same 5 % of total power step size.

During a pre-measurement the *THC* is recorded for each investigated power level. The final measurement for all harmonics shall be carried out at the power level with the highest *THC* value. The type of equipment behavior is quasi-stationary and the final measurement time is started after steady-state is reached, that is about 30 s after changing the settings.

Instantaneous water heaters containing only heating elements that are not supplied through any non-linear electronic device are deemed to comply without testing.

Annex B (informative)

Illustration of limits for harmonic currents

The individual admissible harmonic currents increase linearly with increasing R_{sce} between the minimum value of R_{sce} (33) and the maximum value of R_{sce} found in Tables 2, 3, 4 and 5. The principle is illustrated in Figure B.1 for the 5th harmonic current.

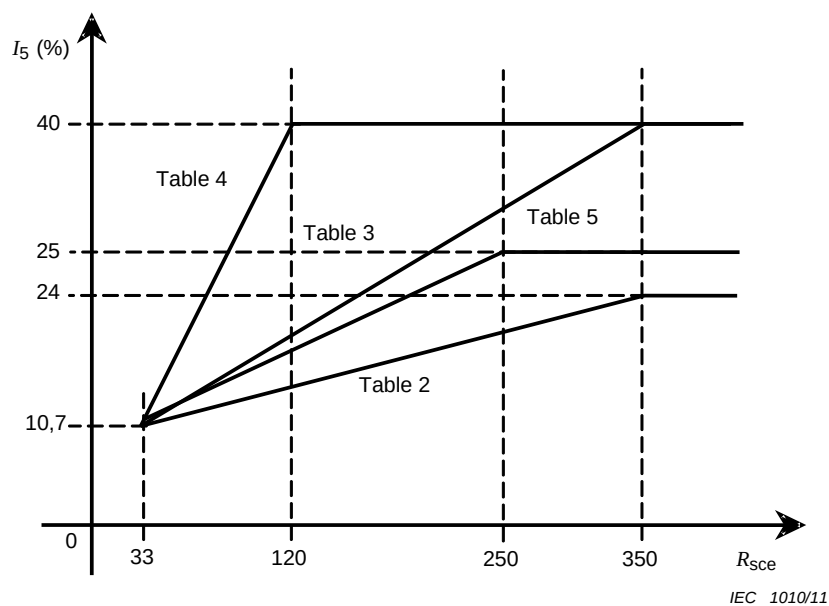


Figure B.1 – Limits of the 5th harmonic current as functions of R_{sce}